Food – Body Connection

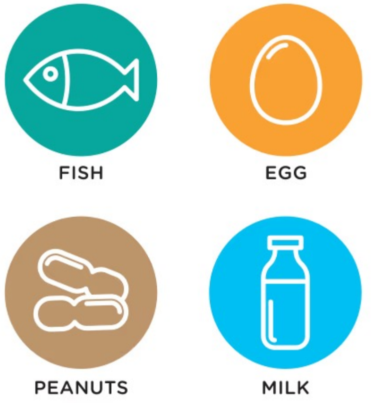
Target audience

- Health practitioners to provide app to patients
  - Dietician
  - Naturopath
  - Herbalist
  - Doctor

- Any individual with symptoms related to food/allergens and unable to pin down causes

Goals

- Design a secure web based app
  - User can register/login



  - User can log:
    - allergens exposed to
    - symptoms
  - User can extract meaningful information from logged data

System Architecture Overview

Frontend is a static web app (GitHub):

- Send requests to an API
- Render responses

Backend (FastAPI):

- Handles authentication
- Stores and analyses data

Database (AWS RDS):

- Stores structured health data (PostgreSQL)
- Is never accessed directly by the frontend

SQL Database

Users
User ID
Email
Password Hash
Created At

Symptoms
Symptom ID
User ID
Symptom Name
Symptom Group

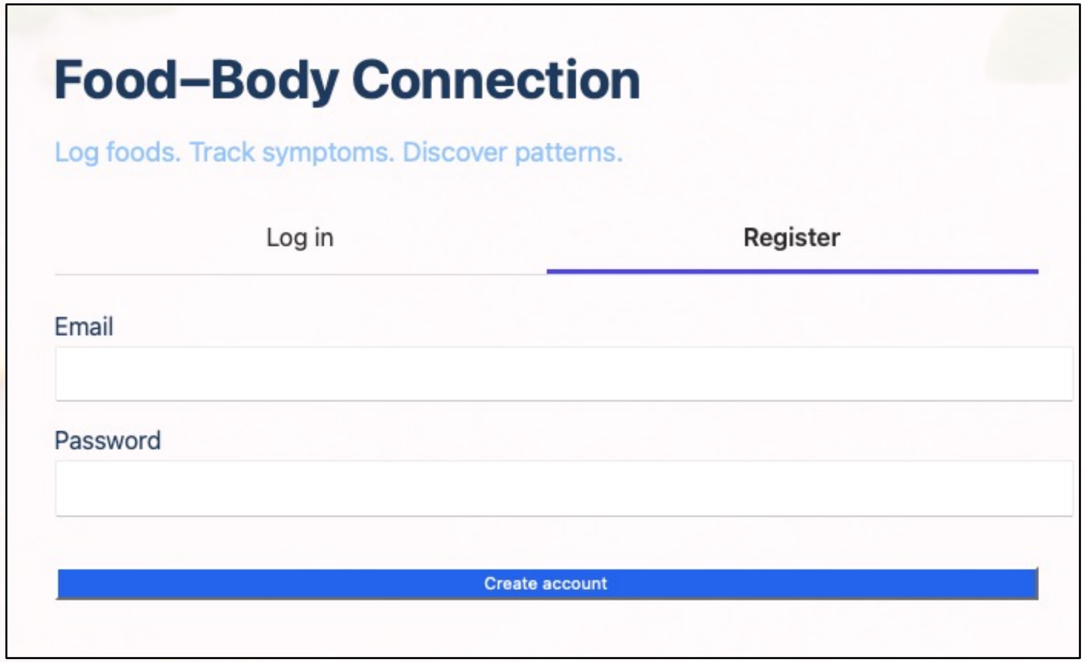
Symptom Log
Symptom Log ID
User ID
Date / Time
Symptom ID
Symptom Intensity

Allergen Log
Allergen Log ID
User ID
Date / Time
Allergen ID
Quantity
Unit ID

Allergens
Allergen ID
User ID
Allergen Name

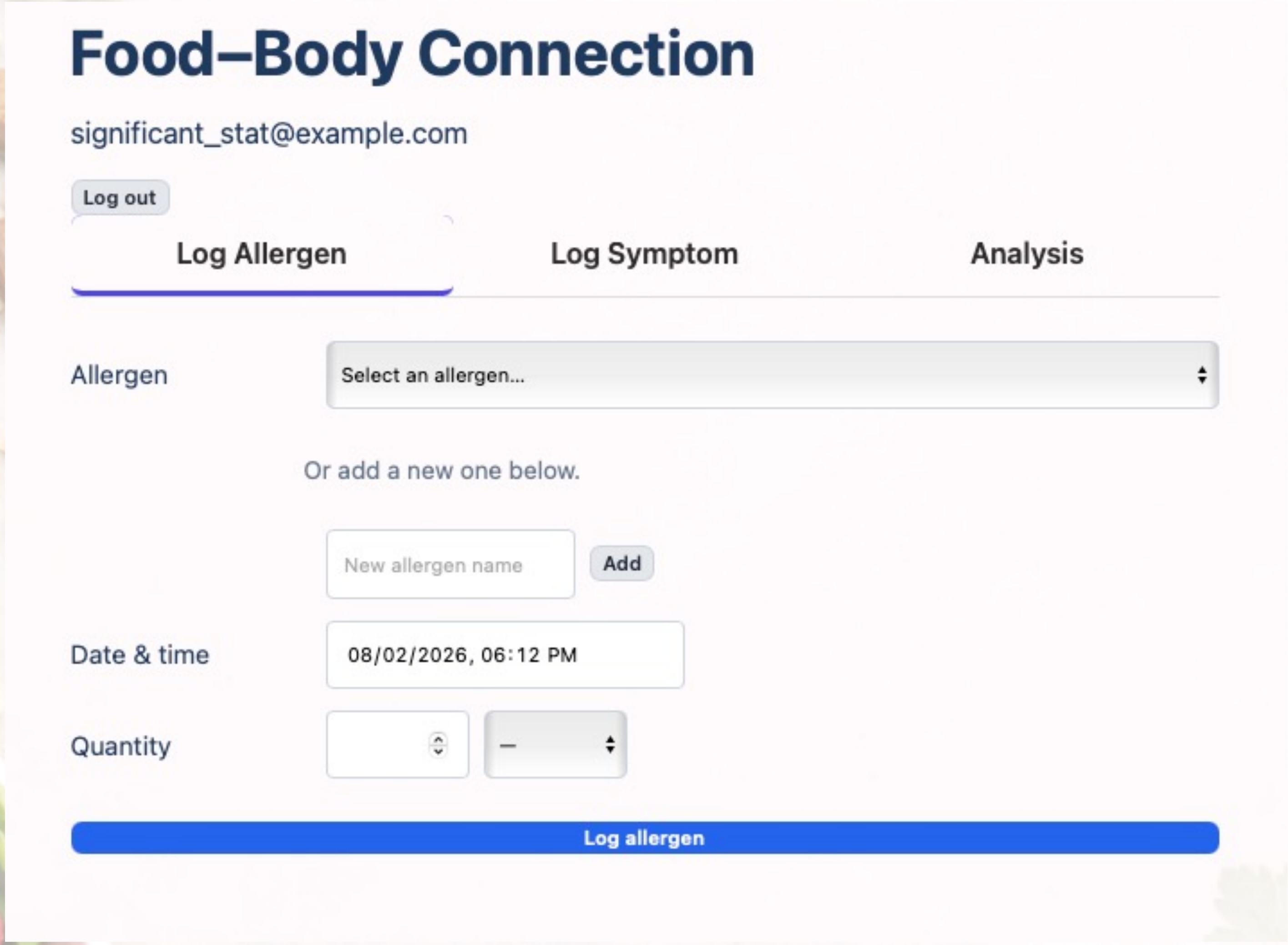
Units
Unit ID
Unit Name
Unit Conversion

Logging in



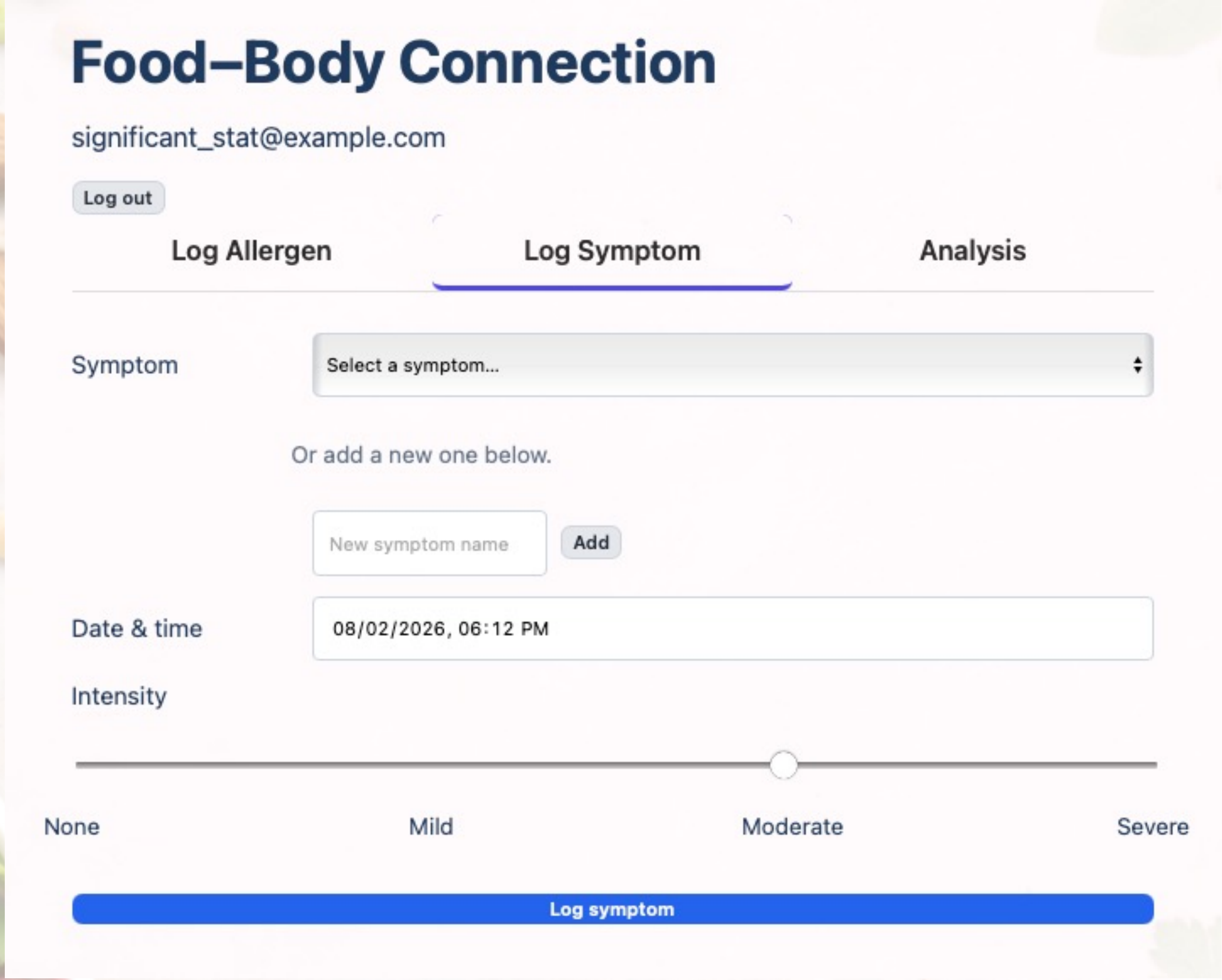
Users
User ID
Email
Password Hash
Created At

Logging an exposure to an allergen



Allergen Log
Allergen Log ID
User ID
Date / Time
Allergen ID
Quantity
Unit ID

Logging a symptom



Symptom Log
Symptom Log ID
User ID
Date / Time
Symptom ID
Symptom Intensity

Analysis - Goals

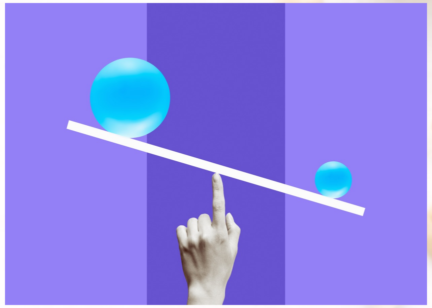
- User to gain a sense of their data

- User to discover most likely allergen triggering symptoms

- User to dive deeper into a particular allergen, time window and grouped symptoms
    - Risk assessment of allergen
    - Time series visual aid
    - Dose – response analysis

Analysis - Challenges

- Small Sample Size
  - 3 years of daily symptoms ≈1000 samples



- Class imbalance
  - No. allergen logs >> No. symptom logs
- Temporal uncertainty
  - Symptoms occurs hours or days after exposure

- Allergens often co-occur
  - Pie = dairy and gluten

- Relies on user entering quality data
- Non stationarity
  - User sensitivity changes over time

Model choice

Model	Binary Outcome	Interpretable	Stable & Reproducible	Handles Correlated Exposures	Fast	Overall Fit
Logistic Regression (L1/L2)	✅	✅✅	✅✅	✅	✅✅	✅✅✅ BEST
Linear Probability Model	✅	✅✅	✅✅	⚠️	✅✅	❌
Bernoulli Naive Bayes	✅	⚠️	✅✅	❌	✅✅	❌
Fisher Exact Test	✅	✅	✅✅	❌	✅✅	❌
Single Decision Tree	✅	⚠️	❌	✅	✅	❌
Random Forest	✅	❌	❌	✅	❌	❌
Gradient Boosting	✅	❌	❌	✅	❌	❌
Bayesian Logistic Regression	✅	✅	⚠️	✅	❌	❌
Kernel SVM (RBF)	✅	❌	❌	✅	❌	❌
Neural Networks	✅	❌	❌	✅	❌	❌

Linking allergens and symptoms

Features

Allergens

Target

Symptoms present

Modelling Techniques

Regularisation (L1 or L2) – mitigates collinearity and overfitting
Nested cross validation – unbiased and stable performance
Class weighting – accounts for class imbalance

Logistic Regression Model

Coefficients

Odds Ratio

Metrics

AUC ROC
Recall
Sample size
Confidence intervals
Collinearity warning

Deep Dive into Modelling Technique

Goal

Identify allergens most strongly associated with symptom occurrence

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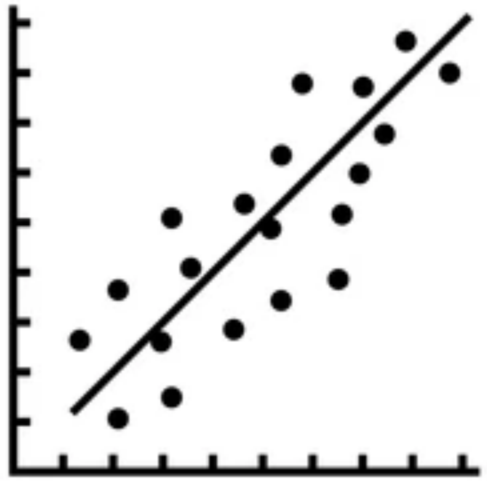
allergens with the largest positive logistic regression coefficients (highest odds ratios)

Deep Dive into Modelling Technique

Model selection & fitting

- Evaluate multiple **exposure–symptom time windows** to account for delayed responses
- Assess **class imbalance** using the proportion of symptom events
- Fit a **regularized logistic regression** model:
  - Penalties: L1 and L2
  - Regularization strength: $C \in \{0.1, 1, 10\}$
- Select the best model using **AUC ROC** to evaluate
- Use **5-fold cross-validation** to estimate:
  - Mean AUC ROC (discrimination)
  - Mean recall (sensitivity to symptom events)

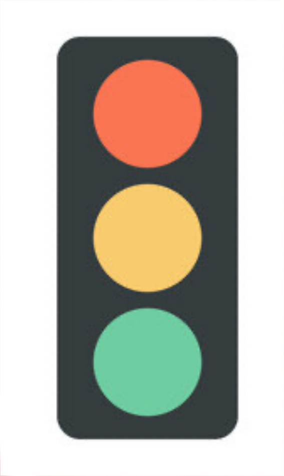


Deep Dive into Modelling Technique

Outputs

- **Odds ratios** for each allergen ($OR=e^{\beta}$)
  identifying allergens with the highest odds of symptom occurrence

- **Model reliability metrics**:
  - AUC ROC
  - Recall
  - Sample size
  - Collinearity indicator
  - Confidence intervals for odds ratios using bootstrap resampling



Validation: Synthetic User and Data

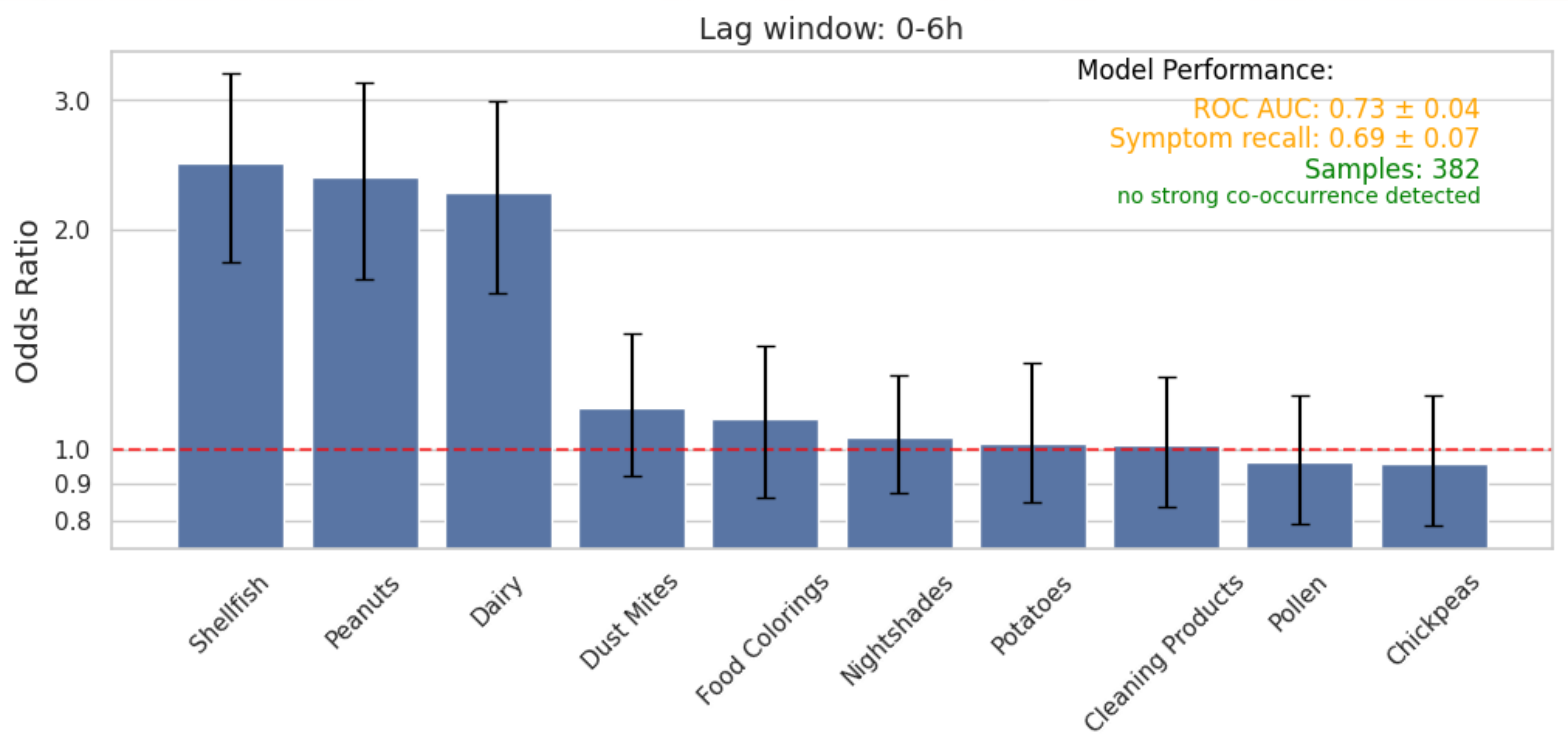
Goal: Evaluate allergen–symptom models using data with **known causal relationships**.

Causal allergens: Dairy, Peanuts, Shellfish

Intensity: Peanuts include a **dose–response effect**

Tests whether models detect real associations

Validation: Synthetic User and Data



Exploring the App Further

food_body_connection

Summary: Food – Body Connection App

Target audience:

- Health practitioners and individuals with symptoms linked to food sensitivities

User Benefit:

- Makes hidden food–body connections visible
- Empowers users to make informed dietary and lifestyle choices

Key Features:

- **Track exposures and symptoms**
- **Analyze relationships:** Identify correlations between exposures and symptoms.
- **Insights & reports:** Odds ratios, intensity charts, and personalized summaries.

Architecture:

- **Frontend:** Static web app (GitHub Pages) → sends requests to API
- **Backend:** FastAPI → handles authentication, data storage, analysis
- **Database:** PostgreSQL on AWS RDS → structured health data

Future Work

- Analysis:
    - Test for regular cycles in symptom occurrence
    - Introduce a new table – food, which automatically allocates allergens
    - Option to email reports
    - Another form displaying raw data users can edit / delete logs
    - Introduce analysis on selected dates
    - Allow users to log multiple allergens / symptoms per event
- Architecture
    - Move to alternative system – long term

https://alkohout.github.io/food_body_connection/dashboard.html

https://github.com/alkohout/food_body_connection

https://www.linkedin.com/in/alisonkohout/